OPTIMAL EXPONENTS FOR THE ŁOJASIEWICZ INEQUALITY OF CUBIC POLYNOMIALS

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Abstract. In this note, we give a completely elementary proof of the Łojasiewicz gradient inequality $|F|^{2\beta} \leq C\,|\nabla F|^2$ for real cubic polynomials in two variables near a critical point. Moreover, we obtain explicit and improved exponents depending on the degeneracy of the quadratic part of F. Our argument relies only on direct estimates, region decompositions, and homogeneous scaling.

1. Introduction

Gradient flows associated with smooth functions play a fundamental role in analysis and dynamical systems. Given a smooth function $F : \mathbb{R}^n \to \mathbb{R}$, the associated gradient flow is defined by the autonomous ODE

$$\gamma'(t) = -\nabla F(\gamma(t)). \tag{1}$$

Along such a trajectory, the energy is monotonically decreasing, since

$$\frac{d}{dt}F(\gamma(t)) = -|\nabla F(\gamma(t))|^2 \leq 0. \tag{2}$$

Thus, the gradient flow provides a natural mechanism for seeking critical points of F.

For a proper smooth function F, the value $F(\gamma(t))$ converges to a critical value as $t \to \infty$. However, this does not imply the convergence of $\gamma(t)$. A fundamental question in the theory of gradient flows is whether the trajectory $\gamma(t)$ itself converges to a critical point.

The resolution of this question relies on a key analytical tool: the Łojasiewicz inequality [Ł84]. Let p be a critical point of F, that is $\nabla F(p) = 0$. The Łojasiewicz inequality [Ł84] asserts that in a neighborhood of p,

$$|F(q) - F(p)|^{2\beta} \leq C|\nabla F(q)|^2, \tag{3}$$

for some exponent $\beta \in \left(\frac{1}{2}, 1\right)$.

This inequality plays an important role in proving the convergence of gradient flows near critical points and also provides quantitative information about the convergence rate. Łojasiewicz inequality also has wide application in other fields such as PDE and geometry, as pioneered by [Sim83].

It is known that the Łojasiewicz inequality holds for real analytic functions, but it may fail for smooth functions in general. Moreover, there exist

proper smooth functions whose gradient flows do not converge. Thus, understanding the structure of the inequality and, in particular, determining the optimal exponent β, becomes a delicate problem [Sim83].

While the inequality appears elementary in form, its proof relies on substantial algebraic tools in the general setting. Furthermore, the optimal exponent β is typically difficult, if not impossible to determine explicitly. The exponent governs the strength of the gradient control near critical points and influences the convergence behavior of gradient trajectories [Sim83].

In this note, we study a simplified model problem in dimension $n = 2$, where F is a polynomial of degree at most 3 and the critical point is located at the origin. Our aim is twofold:

- to prove the Łojasiewicz inequality in this special case using elementary methods;
- to determine explicitly the optimal exponent β.

The main result of this paper provides a complete classification of the optimal exponent according to the degeneracy of the quadratic part.

Theorem 1.1. *Let $F : \mathbb{R}^2 \to \mathbb{R}$ be a polynomial of degree at most 3 satisfying $F(0,0) = 0$ and $\nabla F(0,0) = 0$. Then there exist constants $\delta > 0$ and $C > 0$ such that*

$$|F(x,y)|^{2\beta} \leq C|\nabla F(x,y)|^2 \quad \text{for all } (x,y) \in B_\delta(0), \tag{4}$$

where the optimal exponent β is given by $\frac{2}{3}$.

The remainder of the paper is organized as follows. In Section 2, we explain how the Łojasiewicz inequality implies convergence of gradient flows. In Section 3, we fix notation and basic reductions. In Sections 4–6, we analyze separately the different degeneracy cases of the quadratic part and determine the corresponding exponents β. In Section 7, we prove the optimality of the exponent β.

2. Convergence of gradient flow

In this section, we explain how the Łojasiewicz inequality can imply the convergence of gradient flow of F. To this end, let us assume that the Łojasiewicz inequality (4) holds.

Then let us assume that the gradient flow starts from $t = 1$ and abbreviate

$$F(t) = F(\gamma(t)) \tag{5}$$

We aim to prove $\gamma(t)$ converges as $t \to \infty$. This is equivalent to

$$\int_1^\infty |\gamma'(t)| < \infty \tag{6}$$

By the Łojasiewicz inequality (4), we obtain

$$F'(t) = \frac{d}{dt}F(\gamma(t)) = \langle \nabla F, \gamma'(t)\rangle = -|\nabla F|^2 \leq 0 \tag{7}$$

This implies that $F(t)$ is monotone decreasing, converging to 0. Then $F(t) \geq 0$. Then by (4), there is a $C > 0$ such that

Assume that near a critical point we have

$$|F(q) - F(p)|^{2\beta} \leq C|\nabla F(q)|^2, \qquad \beta \in \left(\frac{1}{2}, 1\right). \tag{8}$$

Let γ be a gradient flow trajectory starting at $t = 1$ and set

$$F(t) := F(\gamma(t)) \geq 0. \tag{9}$$

Along γ we have

$$F'(t) = \frac{d}{dt}F(\gamma(t)) = \langle \nabla F(\gamma(t)), \gamma'(t)\rangle = -|\nabla F(\gamma(t))|^2 \leq 0. \tag{10}$$

By (8) and (10), there exists $c_0 > 0$ such that

$$F'(t) = -|\nabla F(\gamma(t))|^2 \leq -c_0\, F(t)^{2\beta}. \tag{11}$$

Let $\alpha := 2\beta - 1 > 0$. Multiplying (11) by $F(t)^{-2\beta}$ gives

$$\left(F(t)^{-\alpha}\right)' = -\alpha F'(t) F(t)^{-2\beta} \geq \alpha c_0.$$

Integrating from 1 to t yields

$$F(t)^{-\alpha} \geq F(1)^{-\alpha} + \alpha c_0 (t-1) \geq c_1 t$$

for some $c_1 > 0$ and all $t \geq 1$. Equivalently,

$$F(t) \leq C\, t^{-1/(2\beta-1)}. \tag{12}$$

To prove $\gamma(t)$ converges, it suffices to show

$$\int_1^\infty |\gamma'(t)|\, dt < \infty. \tag{13}$$

Since $|\gamma'(t)| = |\nabla F(\gamma(t))| = \sqrt{-F'(t)}$, we apply Cauchy–Schwarz: for any $\varepsilon > 0$,

$$\begin{aligned}
\int_1^\infty |\gamma'(t)|\, dt &= \int_1^\infty \sqrt{-F'(t)}\, dt \\
&\leq \left(\int_1^\infty (-F'(t))\, t^{1+\varepsilon}\, dt\right)^{1/2} \left(\int_1^\infty t^{-1-\varepsilon}\, dt\right)^{1/2} \\
&= \left(\int_1^\infty (-F'(t))\, t^{1+\varepsilon}\, dt\right)^{1/2} \left(\frac{1}{\varepsilon}\right)^{1/2}. \qquad (14)
\end{aligned}$$

Integrating by parts,

$$\int_1^\infty (-F'(t))\, t^{1+\varepsilon}\, dt = \Big[-F(t)t^{1+\varepsilon}\Big]_1^\infty + \int_1^\infty (1+\varepsilon)F(t)\, t^{\varepsilon}\, dt. \tag{15}$$

Choose $\varepsilon > 0$ such that

$$1 + \varepsilon - \frac{1}{2\beta - 1} < 0 \quad \Longleftrightarrow \quad 0 < \varepsilon < \frac{1}{2\beta - 1} - 1, \tag{16}$$

which is possible since $\beta \in (1/2, 1)$. Then by (12),

$$F(t)t^{1+\varepsilon} \leq C\, t^{1+\varepsilon-1/(2\beta-1)} \xrightarrow[t\to\infty]{} 0,$$

hence $\left[-F(t)t^{1+\varepsilon}\right]_1^\infty = F(1) < \infty$.

Moreover, using (12) again,

$$\int_1^\infty F(t)t^\varepsilon\, dt \leq C \int_1^\infty t^{-1/(2\beta-1)+\varepsilon}\, dt.$$

By (16), the exponent satisfies $-1/(2\beta-1)+\varepsilon < -1$, so the integral converges and

$$\int_1^\infty t^{-1/(2\beta-1)+\varepsilon}\, dt = \frac{1}{\frac{1}{2\beta-1} - \varepsilon - 1} < \infty.$$

Therefore the right-hand side of (15) is finite, hence $\int_1^\infty (-F'(t))t^{1+\varepsilon} dt < \infty$. Plugging this into (14) proves (13), and thus $\gamma(t)$ converges as $t \to \infty$.

3. Notation and convention

For two functions f, g, we write $f \lesssim g$, if there exists a constant $C > 0$ independent of the variable such that $f \leq Cg$. Similar definition for $\gtrsim$.

We write $f \approx g$ if $f \lesssim g$ and $f \gtrsim g$. In other words, there exists constant $C > 1$ such that $C^{-1}g \leq f \leq Cg$.

4. Proof of main results

In this section we prove Theorem 1.1. Recall that our function is

$$F(x, y) = ax^2 + by^2 + cx^3 + dx^2y + exy^2 + fy^3.$$

The proof is organized according to the degeneracy of the quadratic part of F near the origin. When the quadratic part is nondegenerate, the behavior of F is essentially governed by a Morse-type structure. When degeneracy occurs, higher-order terms influence the optimal Łojasiewicz exponent and require a more refined analysis.

More precisely, we consider the following three situations: Case A:$a \neq 0$ and $b \neq 0$ (nondegenerate quadratic part); Case B: $a \neq 0$ and $b = 0$ (degeneracy in one direction); Case C: $a = b = 0$ (purely cubic case).

In each case we estimate $|F|$ and $|\nabla F|$ near the origin and determine the optimal exponent β for which the Łojasiewicz gradient inequality holds.

4.1. **Case A.** Assume $a \neq 0$ and $b \neq 0$. Recall that in this case

$$F(x, y) = ax^2 + by^2 + cx^3 + dx^2y + exy^2 + fy^3.$$

We decompose F into its quadratic and cubic parts,

$$Q(x, y) = ax^2 + by^2, \qquad R(x, y) = cx^3 + dx^2y + exy^2 + fy^3,$$

so that $F = Q + R$.

Since $a, b \neq 0$, the quadratic form

$$Q(x, y) = ax^2 + by^2$$

is nondegenerate. Consider the unit circle

$$S^1 = \{(x, y) : x^2 + y^2 = 1\}.$$

The function $(x, y) \mapsto |Q(x, y)|$ is continuous on S^1 and never vanishes. Hence it attains positive minimum and finite maximum values on S^1. By homogeneity of degree two, there exist constants $m, M > 0$ such that

$$m(x^2 + y^2) \leq |Q(x, y)| \leq M(x^2 + y^2)$$

in a sufficiently small neighborhood of the origin. In particular,

$$|Q(x, y)| \approx x^2 + y^2.$$

To estimate F, it remains to control the cubic remainder R. We now bound the higher-order terms. Each monomial in R satisfies

$$|x|^3,\ |x|^2|y|,\ |x||y|^2,\ |y|^3 \lesssim (x^2 + y^2)^{3/2}.$$

Therefore

$$|R(x, y)| \lesssim (x^2 + y^2)^{3/2}.$$

Since $(x^2 + y^2)^{3/2} \leq C(x^2 + y^2)$ for (x, y) sufficiently close to the origin,

$$|R(x, y)| \lesssim x^2 + y^2.$$

Combining the two estimates yields

$$|F(x, y)| \leq |Q(x, y)| + |R(x, y)| \lesssim x^2 + y^2.$$

We now turn to the gradient. We compute

$$F_x = 2ax + 3cx^2 + 2dxy + ey^2, \qquad F_y = 2by + dx^2 + 2exy + 3fy^2.$$

The linear terms dominate near the origin, since $a, b \neq 0$, we have:

$$|2ax| \approx |x|, \qquad |2by| \approx |y|,$$

while the remaining quadratic terms satisfy

$$|3cx^2 + 2dxy + ey^2| \lesssim x^2 + y^2,$$
$$|dx^2 + 2exy + 3fy^2| \lesssim x^2 + y^2.$$

Hence, for (x, y) sufficiently small, we can divide into two cases, $|x| \leq |y|$ and $|x| \geq |y|$.

When $|x| \geq |y|$, we have $F_x^2 \approx x^2 \approx x^2 + y^2$ and $F_y^2 \lesssim x^2 + y^2$.

When $|x| \leq |y|$, we have $F_y^2 \approx y^2 \approx x^2 + y^2$ and $F_x^2 \lesssim x^2 + y^2$.

In either case

$$|\nabla F(x, y)|^2 \approx x^2 + y^2.$$

We now compare the two sides of the Łojasiewicz inequality. Since

$$|F(x, y)|^{2\beta} \approx (x^2 + y^2)^{2\beta}, \qquad |\nabla F(x, y)|^2 \approx x^2 + y^2,$$

the inequality

$$|F|^{2\beta} \leq C|\nabla F|^2$$

is equivalent to

$$(x^2 + y^2)^{2\beta} \lesssim x^2 + y^2,$$

which holds if and only if

$$2\beta \geq 1.$$

Finally, along the line $y = 0$ we have

$$F(x,0) \approx x^2, \qquad |\nabla F(x,0)|^2 \approx x^2,$$

so the same condition $2\beta \geq 1$ is necessary. Therefore the optimal exponent in this case is $\beta = \frac{1}{2}$.

4.2. **Case B.** $a \neq 0, b = 0$. In this case we can write F as:

$$F(x,y) = ax^2 + cx^3 + dx^2y + exy^2 + fy^3, \qquad a \neq 0.$$

Proposition 4.1. *When $a \neq 0, b = 0$, in a sufficiently small neighborhood of $(0,0)$, there exists $C > 0$ such that*

$$|F(x,y)|^{2\beta} \leq C|\nabla F(x,y)|^2$$

holds if

$$\beta \geq \begin{cases} \frac{2}{3}, & f \neq 0, \\ \frac{1}{2}, & f = 0. \end{cases}$$

Proof. We work in a sufficiently small neighborhood of the origin. The partial derivatives are

$$F_x = 2ax + 3cx^2 + 2dxy + ey^2, \qquad F_y = dx^2 + 2exy + 3fy^2,$$

and

$$|\nabla F|^2 = F_x^2 + F_y^2 \geq F_x^2, \qquad |\nabla F|^2 \geq F_y^2.$$

Case 1: $f \neq 0$**.** Define the region decomposition by the comparability boundary $x^2 \approx y^3$:

$$\Omega_1 = \{|x| \leq |y|^{3/2}\}, \qquad \Omega_2 = \{|x| \geq |y|^{3/2}\}.$$

Region Ω_1. By the definition of Ω_1 we have $|x| \leq |y|^{3/2}$. Then

$$|x|^2 \leq |y|^3, \quad |x|^3 \leq |y|^{9/2}, \quad |x|^2|y| \leq |y|^4, \quad |x||y|^2 \leq |y|^{7/2}.$$

Hence

$$|F| \leq |a||x|^2 + |c||x|^3 + |d||x|^2|y| + |e||x||y|^2 + |f||y|^3 \lesssim |y|^3.$$

Next,

$$F_y = 3fy^2 + dx^2 + 2exy.$$

By the triangle inequality and the above bounds,

$$|F_y| \geq 3|f||y|^2 - |d||x|^2 - 2|e||x||y| \geq 3|f||y|^2 - |d||y|^3 - 2|e||y|^{5/2}.$$

For $|y|$ sufficiently small, the terms $|y|^3$ and $|y|^{5/2}$ can be absorbed by $|y|^2$, hence

$$|F_y| \gtrsim |y|^2, \qquad |\nabla F|^2 \geq F_y^2 \gtrsim |y|^4.$$

Therefore

$$|F|^{2\beta} \lesssim |y|^{6\beta}.$$

To have $|y|^{6\beta} \lesssim |y|^4$, we must require

$$6\beta \geq 4 \quad \Longrightarrow \quad \beta \geq \frac{2}{3}.$$

Region Ω_2. By definition $|x| \geq |y|^{3/2}$. Then

$$|y| \leq |x|^{2/3}.$$

In particular, in a small neighborhood where $|x| \leq 1$,

$$|y|^2 \leq |x|^{4/3} \leq |x|, \qquad |x||y| \leq |x|^{5/3}, \qquad |x|^2 \leq |x|.$$

1. Upper bound for $|F|$.
   Since $|y|^3 \leq |x|^2$ under $|y| \leq |x|^{2/3}$, we have

$$|F| \leq C\big(|x|^2 + |y|^3\big) \lesssim |x|^2.$$

2. Lower bound via F_x. Write

$$F_x = 2ax + \big(3cx^2 + 2dxy + ey^2\big).$$

Using the relations above,

$$|3cx^2+2dxy+ey^2| \leq 3|c||x|^2+2|d||x||y|+|e||y|^2 \leq 3|c||x|^2+2|d||x|^{5/3}+|e||x|^{4/3}.$$

For $|x|$ sufficiently small, the terms $|x|^2, |x|^{5/3}, |x|^{4/3}$ are all absorbed by $|x|$; hence we may choose the neighborhood such that

$$3|c||x|^2 + 2|d||x|^{5/3} + |e||x|^{4/3} \leq |a||x|.$$

Therefore

$$|F_x| \geq 2|a||x| - |a||x| = |a||x| \gtrsim |x|, \qquad |\nabla F|^2 \geq F_x^2 \gtrsim |x|^2.$$

3. Exponent comparison. Since $|F| \lesssim |x|^2$,

$$|F|^{2\beta} \lesssim |x|^{4\beta} \lesssim |x|^2 \quad \Longrightarrow \quad 4\beta \geq 2 \Longrightarrow \beta \geq \frac{1}{2}.$$

Combining Ω_1 and Ω_2, Case 1 requires

$$\beta \geq \frac{2}{3}.$$

To see optimality when f = 0, we argue as follows. Along $x = 0$,

$$F(0, y) = fy^3, \qquad |\nabla F(0, y)| \approx |y|^2,$$

hence $|y|^{6\beta} \lesssim |y|^4$ forces $\beta \geq \frac{2}{3}$. Thus

$$\beta_{\text{opt}} = \frac{2}{3} \qquad (f \neq 0).$$

Case 2: $f = 0$**.** Now

$$F = ax^2 + cx^3 + dx^2y + exy^2.$$

We further split into $e \neq 0$ and $e = 0$.

Case 2a: $f = 0,\ e \neq 0$. Define the region decomposition:

$$\Lambda_1 = \{|x| \leq |y|^2\}, \qquad \Lambda_2 = \{|x| \geq |y|^2\}.$$

Region Λ_2. Assume $|x| \geq |y|^2$. Then $|y|^2 \leq |x|$, and in a small neighborhood we also have $|x|^2 \leq |x|$. Hence

$$|F| \lesssim |x|^2.$$

Moreover,

$$|F_x| \geq 2|a||x| - 3|c||x|^2 - 2|d||x||y| - |e||y|^2.$$

Under $|y|^2 \leq |x|$, we have $|x||y| \leq |x| \cdot |x|^{1/2} = |x|^{3/2}$, which is higher order than $|x|$, and $|x|^2$ is also higher order than $|x|$. Also $|y|^2 \leq |x|$. Therefore

$$|F_x| \geq 2|a||x| - C|x| \gtrsim |x|$$

for $|x|$ sufficiently small, hence

$$|\nabla F|^2 \geq F_x^2 \gtrsim |x|^2.$$

Thus

$$|F|^{2\beta} \lesssim |x|^{4\beta} \lesssim |x|^2 \quad \Longrightarrow \quad 4\beta \geq 2 \Longrightarrow \beta \geq \frac{1}{2}.$$

Region Λ_1. Assume $|x| \leq |y|^2$. Then

$$|x|^2 \leq |y|^4, \qquad |x||y| \leq |y|^3,$$

so

$$|F| \lesssim |y|^4.$$

Write

$$F_x = (2ax + ey^2) + (3cx^2 + 2dxy).$$

Moreover, using $|x|^2 \leq |y|^4$ and $|x||y| \leq |y|^3$,

$$|3cx^2 + 2dxy| \lesssim |y|^4 + |y|^3.$$

We further split Λ_1 into two subregions.

- If

$$|2ax + ey^2| \geq \frac{|e|}{2}|y|^2,$$

then

$$|F_x| \geq \frac{|e|}{2}|y|^2 - C|y|^3 \gtrsim |y|^2$$

for $|y|$ sufficiently small, hence

$$|\nabla F|^2 \geq F_x^2 \gtrsim |y|^4.$$

Since $|F| \lesssim |y|^4$, we get

$$|F|^{2\beta} \lesssim |y|^{8\beta} \lesssim |y|^4 \quad \Longrightarrow \quad 8\beta \geq 4 \Longrightarrow \beta \geq \frac{1}{2}.$$

- If

$$|2ax + ey^2| \leq \frac{|e|}{2}|y|^2,$$

then

$$|2ax| \geq \frac{|e|}{2}|y|^2 \quad \Longrightarrow \quad |x| \gtrsim |y|^2.$$

Together with $|x| \le |y|^2$, we obtain

$$|x| \approx |y|^2.$$

In particular, $|F_x| \gtrsim |x| \approx |y|^2$, hence

$$|\nabla F|^2 \ge F_x^2 \gtrsim |y|^4,$$

and again $\beta \ge \frac{1}{2}$ follows.

Therefore, in Case 2a we conclude

$$\beta \ge \frac{1}{2}.$$

Necessity follows from the path $y = 0$ (see below); hence,

$$\beta_{\text{opt}} = \frac{1}{2} \qquad (f = 0,\ e \ne 0).$$

Case 2b: $f = 0,\ e = 0$. Now

$$F = ax^2 + cx^3 + dx^2y = x^2(a + cx + dy).$$

Define

$$\Gamma_1 = \left\{ |cx + dy| \le \frac{|a|}{2} \right\}, \qquad \Gamma_2 = \left\{ |cx + dy| \ge \frac{|a|}{2} \right\}.$$

Also,

$$F_x = 2ax + 3cx^2 + 2dxy, \qquad F_y = dx^2.$$

Region Γ_1. Since $|a + cx + dy| \ge |a| - |cx + dy| \ge \frac{|a|}{2}$, we have

$$|F| = |x|^2|a + cx + dy| \gtrsim |x|^2.$$

Moreover,

$$|F_x| \ge 2|a||x| - 3|c||x|^2 - 2|d||x||y|.$$

In a sufficiently small neighborhood, $|x|^2$ and $|x||y|$ are higher order than $|x|$ and can be absorbed, hence

$$|F_x| \gtrsim |x| \quad \Longrightarrow \quad |\nabla F|^2 \ge F_x^2 \gtrsim |x|^2.$$

Therefore, $|F|^{2\beta} \lesssim |\nabla F|^2$ forces

$$|x|^{4\beta} \lesssim |x|^2 \quad \Longrightarrow \quad 4\beta \ge 2 \Longrightarrow \beta \ge \frac{1}{2}.$$

Region Γ_2. We have

$$|F| = |x|^2|a + cx + dy| \le |x|^2\big(|a| + |c||x| + |d||y|\big) \lesssim |x|^2$$

in a small neighborhood. The same lower bound $|\nabla F|^2 \gtrsim |x|^2$ holds as above, hence again $\beta \ge \frac{1}{2}$.

To see optimality when Optimality for $f = 0$. we argue as follows Along $y = 0$,

$$F(x, 0) = ax^2 + cx^3 \approx x^2, \qquad |\nabla F(x, 0)|^2 \approx x^2,$$

so $|x|^{4\beta} \lesssim |x|^2$ forces $\beta \ge \frac{1}{2}$. Thus

$$\beta_{\text{opt}} = \frac{1}{2} \qquad (f = 0),$$

and this covers both $e \neq 0$ and $e = 0$.

□

4.3. **Case C.** $a = b = 0$

Recall that in this case

$$F(x, y) = cx^3 + dx^2y + exy^2 + fy^3.$$

We study the Łojasiewicz inequality near $(0, 0)$.

Proposition 4.2. *When $a = b = 0$, in a sufficiently small neighborhood of $(0, 0)$, there exists $C > 0$ such that*

$$|F(x, y)|^{2\beta} \leq C|\nabla F(x, y)|^2$$

The case x=0 can be treated separately and is easier, so we restrict to $x \neq 0$.

When $x \neq 0$, set

$$\lambda = \frac{y}{x}.$$

Then

$$F(x, y) = x^3 Q(\lambda) \tag{17}$$

where

$$Q(\lambda) = c + d\lambda + e\lambda^2 + f\lambda^3.$$

A direct computation gives

$$F_x = x^2 A(\lambda), \qquad F_y = x^2 B(\lambda),$$

where

$$A(\lambda) = 3c + 2d\lambda + e\lambda^2, \tag{18}$$

$$B(\lambda) = d + 2e\lambda + 3f\lambda^2. \tag{19}$$

Hence

$$|\nabla F(x, y)|^2 = F_x^2 + F_y^2 = x^4 G(\lambda), \tag{20}$$

where

$$G(\lambda) = A(\lambda)^2 + B(\lambda)^2. \tag{21}$$

We now distinguish two cases

I. There is no real λ such that $A(\lambda) = B(\lambda) = 0$.

Proof of Proposition 4.2 in Case 0. The assumption in Case 0 states that there is no real λ such that $A(\lambda) = B(\lambda) = 0$. Since

$$G(\lambda) = A(\lambda)^2 + B(\lambda)^2,$$

we have

$$G(\lambda) = 0 \quad \Longleftrightarrow \quad A(\lambda) = B(\lambda) = 0.$$

Hence $G(\lambda) \neq 0$ for all $\lambda \in \mathbb{R}$. Clearly $G(\lambda) \geq 0$, hence

$$G(\lambda) > 0 \quad \forall \lambda \in \mathbb{R}.$$

To study the growth of $G(\lambda)$ as $|\lambda| \to \infty$, we examine its leading term. Since

$$A(\lambda) = 3c + 2d\lambda + e\lambda^2, \qquad B(\lambda) = d + 2e\lambda + 3f\lambda^2,$$

the leading term of $G(\lambda) = A(\lambda)^2 + B(\lambda)^2$ is

$$e^2\lambda^4 + 9f^2\lambda^4 = (e^2 + 9f^2)\lambda^4.$$

Since $|x| + |y| = (1 + |\lambda|)|x|$, from (17) we obtain

$$|F(x,y)|^{4/3} \leq C_2(1 + |\lambda|)^4 x^4 = C_2(1 + |\lambda|)^4 \frac{|\nabla F(x,y)|^2}{G(\lambda)}. \tag{22}$$

Since $G(\lambda) > 0$ for all $\lambda \in \mathbb{R}$, the function

$$H(\lambda) := \frac{(1 + |\lambda|)^4}{G(\lambda)}$$

is continuous on $\mathbb{R}$. Moreover, if $(e, f) \neq (0, 0)$, then

$$G(\lambda) \sim (e^2 + 9f^2)\lambda^4 \quad \text{as } |\lambda| \to \infty,$$

and hence

$$H(\lambda) \to \frac{1}{e^2 + 9f^2} \quad \text{as } |\lambda| \to \infty.$$

Therefore H is bounded on $\mathbb{R}$. Consequently, there exists $C > 0$ such that

$$\frac{(1 + |\lambda|)^4}{G(\lambda)} \leq C \quad \text{for all } \lambda \in \mathbb{R}.$$

Combining this with (22), we obtain

$$|F(x,y)|^{4/3} \leq C|\nabla F(x,y)|^2.$$

Thus Case 0 holds.

□

$$A(\lambda_0) = B(\lambda_0) = 0.$$

Proof. λ_0 is a multiple root of Q.

Since

$$Q'(\lambda) = B(\lambda),$$

and

$$3Q(\lambda) - \lambda Q'(\lambda) = A(\lambda),$$

we obtain

$$Q(\lambda_0) = Q'(\lambda_0) = 0.$$

Thus λ_0 is at least a double root of Q.

Hence either

$$Q(\lambda) = k(\lambda - \lambda_0)^3,$$

or

$$Q(\lambda) = k(\lambda - \lambda_0)^2(\lambda - \mu), \quad \mu \neq \lambda_0.$$

II. Triple root.
If

$$F(x, y) = k(y - \lambda_0 x)^3, (k \neq 0)$$

set $u = y - \lambda_0 x$. Then

$$F = ku^3, \qquad |\nabla F|^2 = 9k^2(\lambda_0^2 + 1)u^4.$$

Since

$$|F|^{4/3} = |k|^{4/3}|u|^4,$$

hence

$$|\nabla F|^2 \approx u^4$$

we obtain

$$|F|^{4/3} \leq C|\nabla F|^2.$$

III. Double root.
Assume

$$F(x, y) = k(y - \lambda_0 x)^2(y - \mu x), \qquad \mu \neq \lambda_0.$$

Set

$$u = y - \lambda_0 x, \qquad v = y - \mu x,$$

so that $F = ku^2 v$.

1. Upper bound. We have

$$|F|^{4/3} = |k|^{4/3}|u|^{8/3}|v|^{4/3}.$$

Divide into two regions

$$\Omega_1 = \{|u| \leq 2|v|\}, \qquad \Omega_2 = \{|u| \geq 2|v|\}.$$

In Ω_1,

$$|u|^{8/3}|v|^{4/3} = (u^2v^2)\,|u|^{2/3}|v|^{-2/3} \leq 2^{2/3}u^2v^2.$$

In Ω_2,

$$|u|^{8/3}|v|^{4/3} = |u|^4\Big(\frac{|v|}{|u|}\Big)^{4/3} \leq 2^{-4/3}u^4.$$

Hence in both regions,

$$|F|^{4/3} \approx u^2v^2 + u^4. \tag{23}$$

2. Lower bound
Using $u_x = -\lambda_0,\ u_y = 1$ and $v_x = -\mu,\ v_y = 1$, we compute

$$F_x = k(2uu_xv + u^2v_x) = k(-2\lambda_0 uv - \mu u^2),$$

$$F_y = k(2uu_yv + u^2v_y) = k(2uv + u^2).$$

So

$$\frac{F_x}{k} = -\lambda_0(2uv) - \mu(u^2), \qquad \frac{F_y}{k} = (2uv) + (u^2). \tag{24}$$

• Multiply the second identity in (24) by μ and add to the first:

$$\mu\frac{F_y}{k} + \frac{F_x}{k} = \mu\big((2uv) + (u^2)\big) - \lambda_0(2uv) - \mu(u^2) = (\mu - \lambda_0)(2uv).$$

Hence

$$2uv = \frac{\mu F_y + F_x}{k(\mu - \lambda_0)}. \tag{25}$$

• Multiply the second identity in (24) by λ_0 and add to the first:

$$\lambda_0\frac{F_y}{k} + \frac{F_x}{k} = \lambda_0\big((2uv) + (u^2)\big) - \lambda_0(2uv) - \mu(u^2) = -(\mu - \lambda_0)u^2.$$

Hence

$$u^2 = -\frac{F_x + \lambda_0 F_y}{k(\mu - \lambda_0)}. \tag{26}$$

From (25) and (26), there exists a constant $C > 0$ (depending only on k, λ_0, μ) such that

$$(2uv)^2 + (u^2)^2 \le C(F_x^2 + F_y^2) = C|\nabla F|^2. \tag{27}$$

Since

$$(2uv)^2 + (u^2)^2 = 4u^2v^2 + u^4 \approx u^2v^2 + u^4, \tag{28}$$

we conclude that

$$u^2v^2 + u^4 \approx |\nabla F|^2. \tag{29}$$

Thus the double root case holds. □

5. Optimality of the exponent

We show that the lower bound $\beta \ge \frac{2}{3}$ is sharp by an example:

Proposition 5.1 (Optimality of the exponent)**.** *The optimal exponent in Łojasiewicz inequality* (4) *for cubic polynomial*

$$F(x, y) = ax^2 + by^2 + cx^3 + dx^2y + exy^2 + fy^3$$

is $\beta = \frac{2}{3}$.

Proof. We have already proved that the inequality

$$|F(x, y)|^{4/3} \le C|\nabla F(x, y)|^2$$

holds near $(0, 0)$ for every cubic polynomial in the form of $F(x, y) = ax^2 + by^2 + cx^3 + dx^2y + exy^2 + fy^3$.

To show optimality, it suffices to exhibit an example for which the inequality fails whenever $\beta < \frac{2}{3}$. We therefore consider the model example

$$F(x, y) = y^3.$$

Along the path $(x, y) = (0, t)$ with $t \to 0$, we have

$$|F| = |t|^3, \qquad |\nabla F|^2 = 9|t|^4.$$

Hence

$$|F|^{2\beta} = |t|^{6\beta}.$$

If the inequality were true for some $\beta < \frac{2}{3}$, then

$$|t|^{6\beta} \leq C|t|^4$$

would hold for small t, which is impossible since $6\beta < 4$. Therefore the inequality fails for every $\beta < \frac{2}{3}$. □

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